

# Naive Bayes Classifier | Part 2: Independent Events in Probability

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## 1 What are Independent Events?

Two events are **independent** if the outcome of one **does not affect** the outcome of the other.

**Example:**

- Tossing two coins.
- Event A = First coin is heads.
- Event B = Second coin is tails.
- These are **independent** because what happens to the first coin does not affect the second.

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## 2 Probability Rule for Independent Events

If A and B are independent:

$$P(A \cap B) = P(A) \cdot P(B)$$

That means the probability of **both happening together** is just the product of their individual probabilities.

**Example:**

- $P(\text{Head on first coin}) = 0.5$
- $P(\text{Tail on second coin}) = 0.5$
- $P(\text{Head and Tail}) = 0.5 \times 0.5 = 0.25$

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## 3 Why is this important for Naive Bayes?

In Naive Bayes, we often deal with **many features** (words in a document, symptoms in medical diagnosis, etc.).

Naive Bayes makes a **naïve assumption**:

-> All features are **conditionally independent** given the class.

So instead of calculating a very complicated joint probability like:

$$P(X_1, X_2, X_3 | \text{Class})$$

We simplify it as:

$$P(X_1 | \text{Class}) \cdot P(X_2 | \text{Class}) \cdot P(X_3 | \text{Class})$$

This assumption makes Naive Bayes **fast, simple, and scalable**, even though in real life features are not always perfectly independent.

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#### 4 Intuition Example

Let's say you're classifying whether a person has flu or not.

- Features: **fever, cough, sore throat**.
- Naive Bayes assumes:
  1. Probability of fever given flu is independent of probability of cough given flu.
  2. Probability of sore throat given flu is independent of both.

So:

$$P(Flu|Symptoms) \propto P(Flu) \cdot P(Fever|Flu) \cdot P(Cough|Flu) \cdot P(SoreThroat|Flu)$$

This independence assumption makes the calculation **much simpler**.

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#### ➤ Quick Recap

- **Independent events** = outcome of one does not affect the other.
  - Rule:  $P(A \cap B) = P(A) \cdot P(B)$ .
  - Naive Bayes assumes **all features are independent given the class**, which is why it's called "naive."
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